Z-test and t-test

Z-test

- A Z-test is a statistical test that uses the standard normal (Z) distribution.
- We generally use a Z-test when the population standard deviation sigma is known.
- Population standard deviation is known.
- The sample is randomly selected, and the relevant assumptions are satisfied.
- It is commonly used for large samples; however, it can also be used for small samples when the population standard deviation is known and the population is normally distributed.

Formula

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Where:

- $\bar{x}$ = sample mean
- μ = population mean
- σ = population standard deviation
- n = sample size

- A hospital knows that the population mean waiting time is 30 minutes, with a population standard deviation of 5 minutes. A sample of 25 patients has a mean waiting time of 32 minutes.
- Given:
- Population mean = 30
- Population standard deviation = 5
- Sample mean = 32
- Sample size = 25

Calculation:

$$Z = \frac{32 - 30}{5/\sqrt{25}}$$

$$Z = \frac{2}{1} = 2$$

The calculated Z-value is **2**.

t-test

- A t-test is a statistical test that uses the Student's t-distribution.
- We generally use a t-test when the population standard deviation sigma is unknown and we estimate it using the sample standard deviation s.
- Population standard deviation is unknown.
- Sample standard deviation is used instead.
- It is particularly useful for small samples when the population is approximately normally distributed.

Formula (one-sample t-test):

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Where:

- $\bar{x}$ = sample mean
- μ = hypothesized population mean
- s = sample standard deviation
- n = sample size

Example:

- A researcher wants to test whether the average patient waiting time differs from 30 minutes. A sample of 10 patients has a mean waiting time of 32 minutes and a sample standard deviation of 4 minutes.
- Given:
- Population mean = 30
- Sample mean = 32
- Sample standard deviation = 4
- Sample size = 10

Calculation:

$$t = \frac{32 - 30}{4/\sqrt{10}}$$

$$t = \frac{2}{1.265} \approx 1.58$$

The calculated t-value is approximately **1.58**, with $10 - 1 = 9$ degrees of freedom.

Conditions for Z-test and t-test

- **Condition 1: Sample size**
- Sample size n >=30: The sampling distribution of the mean is often approximately normal, especially when the population is not extremely skewed and has no severe outliers.
- Sample size n < 30: If the population is approximately normally distributed, we can use a t-test when the population standard deviation is unknown, or a Z-test when it is known.
- Important: A sample size below 30 does not automatically mean we must use a non-parametric test.

- *When do we use a non-parametric test?*
- A non-parametric test may be appropriate when the assumptions of a parametric test are not satisfied, depending on the research question and data type.

Examples for students

Situation	Appropriate test
Sample size 50; population SD known	Z-test, if assumptions hold
Sample size 15; population SD unknown; population approximately normal	t-test
Sample size 12; strongly skewed data with serious outliers	Consider a suitable non-parametric or robust method

What is a right-tail test

A **right-tail test** is a hypothesis test in which we want to check whether a population value is **significantly greater than a particular value**.

For example, a company claims that the average weight of a packet is 500 grams. We want to test whether the actual average weight is **greater than 500 grams**.

- $H_0 : \mu = 500$ — Null hypothesis
- $H_1 : \mu > 500$ — Alternative hypothesis

Because the alternative hypothesis uses the **greater than** ($>$) **sign**, we use a right-tail test.

What is a left-tail test?

A **left-tail test** is a hypothesis test used when we want to check whether a population mean is **significantly less than a particular value**.

For example, a company claims that the average weight of a packet is 500 grams. We want to test whether the actual average weight is **less than 500 grams**.

- Null hypothesis: $H_0 : \mu = 500$
- Alternative hypothesis: $H_1 : \mu < 500$

Because the alternative hypothesis uses the **less-than ($<$) sign**, we use a left-tail test.

What is a two-tail test?

A **two-tail test** is a hypothesis test used when we want to check whether a population mean is **different from a particular value**. It can be either greater than or less than that value.

For example, a company claims that the average weight of a packet is 500 grams. We want to test whether the actual average weight is different from 500 grams.

- Null hypothesis: $H_0 : \mu = 500$
- Alternative hypothesis: $H_1 : \mu \neq 500$

The symbol $\neq$ means **not equal to**.

Because the actual mean could be either greater than or less than 500, we use a two-tail test.